# QUEUING THEORY: PART 2

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# Pure Birth Model

- The population increases over time due to “births”.
- No reduction in the population. Only arrival occurs.
- Example: Bacterial growth in nutrient-rich environment, creation of birth certificates for newly born babies.
- Exponential distribution is used to describe the inter-arrival time in pure birth model.

# Pure Birth Model

Say

$p_0(t)$ = Probability of no arrivals during a period of time $t$

If the inter-arrival time is exponential and the arrival rate is λ customers per unit time, then

$$p_0(t) = P\{\text{interarrival time} \geq t\}$$
$$= 1 - P\{\text{interarrival time} \leq t\}$$
$$= 1 - \left(1 - e^{-\lambda t}\right)$$
$$= e^{-\lambda t}$$

# Pure Birth Model

Say

$p_n(t)$ = Probability of $n$ arrivals during a period of time $t$

If $n$ is the number of arrivals during time period $t$ with mean $1/\lambda$, then $p_n(t)$ is obtained as

$$p_n(t) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}, \quad n = 0, 1, 2, 3.....$$

# Pure Birth Model

**Example:** A customer support center has a phone line where new calls are received at the rate of one call every 4 minutes. The time between phone calls follows an exponential distribution. Calculate

(a) The average number of calls per year.

(b) The probability that the next call arrives within 2 minutes.

(c) The probability that the next call arrives after 5 minutes.

# Pure Birth Model

**Solution:** Arrival time between calls, $T$ = 4 minutes

(a) Calls per day = (24×60)/4 = 360

(b) Given, $t$ = 2 minutes

Arrival rate, $\lambda$ = 1/4 = 0.25 calls per minute

Probability that the next call arrives within 2 minutes

$$P(T \le t) = 1 - e^{-\lambda t}$$

$$= 1 - e^{-0.25 \times 2}$$

$$= 0.393$$

# Pure Birth Model

**Solution:**

(c) Given, $t$ = 4 minutes

Probability that the next call arrives after 5 minutes

$$P(T > t) = e^{-\lambda t}$$

$$= e^{-0.25 \times 5}$$

$$= 0.287$$

Chances of getting a call after 5 minutes is 28.7%.

# Pure Death Model

- This model is the opposite of the Pure Birth Model.
- The system loses members over time, but no new arrivals occur.
- **Examples:** Leaves falling from a tree in autumn season, people leaving a queue when the service counter closes permanently, Battery-powered devices gradually shutting down as batteries die.
- Models decay, failure, or departure
- Exponential distribution is used

# Pure Death Model

- In this model, the system starts with *N* customers at time 0 and no arrivals are allowed.
- Departures occur at the rate *μ* customers per unit time.
- The probability $p_n(t)$ of *n* customers remaining after *t* time units,

$$p_n(t) = \frac{(\mu t)^{N-n} e^{-\mu t}}{(N-n)!}, \quad n = 1, 2, 3....., N$$

$$p_0(t) = 1 - \sum_{n=1}^{N} p_n(t)$$

# General Poisson Queueing Model

- This model assumes that both arrival and departure are **state dependent**.
- It means it is dependent on the number of customers in the service facility.
- For example, the number of machine break-down decreases as the number of broken machine increases. This is because only working machines can generate new breakdowns.
- Define

$n$ = Number of customers in the system (in-queue plus in-service)

$\lambda_n$ = Arrival rate, given $n$ customers in the system

$\mu_n$ = Departure rate, given $n$ customers in the system

$p_n$ = steady-state probability of $n$ customers in the system

# General Poisson Queueing Model

- $p_n$ is obtained as function of $\lambda_n$ and $\mu_n$.
- The probability is used to determine the system's measures of performance such as **average queue length**, **average waiting time** and **average utilization of the facility**.
- **Average utilization of the facility:** Proportion of time the service facility (like a machine or server) is busy serving customers.

$$\text{Average utilization of facility} = \frac{\text{Arrival rate}}{\text{Service rate}}$$

The formula is for single server system

# General Poisson Queueing Model

- The probabilities are determined by using transition-rate diagram

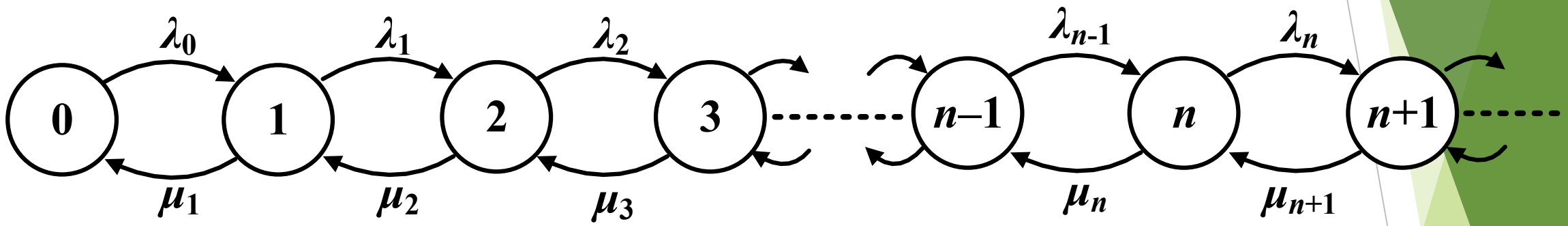



**Figure:** Poisson queues transition diagram

$n$ = Number of customers in the system (in-queue plus in-service)

$\lambda_n$ = Arrival rate, given $n$ customers in the system

$\mu_n$ = Departure rate, given $n$ customers in the system

# General Poisson Queueing Model

- Under steady state conditions, for $n>0$, the expected rate of flow into and out of state $n$ must be equal. The state $n$ can be changed to $n-1$ and $n+1$. Hence,

$$\text{Expected rate of flow into state } n = \lambda_{n-1} p_{n-1} + \mu_{n-1} p_{n+1}$$

And

$$\text{Expected rate of flow out of state } n = (\lambda_n + \mu_n) p_n$$

Equating the both rates, the equation is obtained as

$$\lambda_{n-1} p_{n-1} + \mu_{n-1} p_{n+1} = (\lambda_n + \mu_n) p_n, \quad n = 1, 2, 3, \ldots\ldots$$

# General Poisson Queueing Model

- The equation with $n$ = 0 is

$$\lambda_0 p_0 = \mu_1 p_1$$

- For $n$=1, the equation is

$$\lambda_0 p_0 + \mu_2 p_2 = (\lambda_0 + \mu_1) p_1$$

- The generalized equation is

$$p_n = \left( \frac{\lambda_{n-1}\lambda_{n-2}....\lambda_0}{\mu_n \mu_{n-1....}\mu_1} \right) p_0,\ n = 1,2,3,.....$$

- The value of $p_0$ is determined from $\sum_{n=0}^{\infty} p_n = 1$

# One example

- Let $\lambda_0$=5 customers per hour, $\lambda_1$=4 customers per hour, $\lambda_2$=3 customers per hour; no arrivals take place if there are 3 are more customers in the systems. Service rates: $\mu_1$=1 customer per hour, $\mu_2$=4 customer per hour, $\mu_3$=10 customer per hour. Find out the probability that there will be no customer in the system.

Solution:

$$p_n = \left( \frac{\lambda_{n-1}\lambda_{n-2}....\lambda_0}{\mu_n\mu_{n-1....}\mu_1} \right) p_0,$$

$$p_1 = \frac{\lambda_0}{\mu_1} p_0 = 5p_0; \quad p_2 = \frac{\lambda_0\lambda_1}{\mu_1\mu_2} p_0 = \frac{5\times 4}{1\times 4} p_0 = 5p_0;$$

$$p_3 = \frac{\lambda_0\lambda_1\lambda_2}{\mu_1\mu_2\mu_3} p_0 = \frac{5\times 4\times 3}{1\times 4\times 10} p_0 = 1.5p_0; p_4 = p_5 = \ldots = 0.$$

$$\sum_{n=0}^{\infty} p_n = 1 = p_0 + p_1 + p_2 + p_3 + p_4 + \ldots$$

$$= p_0 + 5p_0 + 5p_0 + 1.5p_0 = 12.5p_0 = 1 \Rightarrow 0.08$$

# Special case: arrival and service rates constant

$$p_n = \left(\frac{\lambda}{\mu}\right)^n p_0 = \rho^n p_0.$$

$$p_0\left(1+\rho+\rho^2+\ldots\right)=1 \Rightarrow p_0 = \left(1-\rho\right)$$

$$L_s = \sum_{n=0}^{\infty} n\rho^n\left(1-\rho\right) = \left(1-\rho\right)\rho\frac{\mathrm{d}}{\mathrm{d}\rho}\sum_{n=0}^{\infty}\rho^n = \frac{\rho}{1-\rho}.$$

# General Poisson Queueing Model

- **Example:** On an average 10 patients keep waiting at the clinic of a doctor. Doctor spends 10 minutes for seeing a patient. On an average 19 patients remain in the system (queue+ consultation) in 2 hours. In two hours, how much time, doctor is busy? What is the arrival rate of customers? What is the average waiting time? Assume Poisson arrival and negative exponential service time.

- **Solution:** **Given,**

**Average number of patients waiting in the queue, $L_q$ = 10**

**Average number of patients in the system (queue + consultation), $L$ = 19 patients**

**Service time, $\mu$ = 10 minutes per patient**

**or 1/10 patients per minute or 6 patients per hour**

# General Poisson Queueing Model

We know,

$$L = \frac{\rho}{1-\rho}$$

$$\Rightarrow \rho = \frac{L}{1+L} = \frac{19}{1+19} = 0.95$$

where $\rho$ is the average utilization of facility

**Busy time of doctor =** $\rho \times 2 = 0.95 \times 2 = 1.9$ hr = 114 min

Using $\rho = \frac{\lambda}{\mu}$ arrival rate is $\lambda = \rho \times \mu = 0.95 \times 6 = 5.7$ patients per hour

Average waiting time,

$$W_q = \frac{L_q}{\lambda} = \frac{10}{5.7} = 1.75 \text{ hr} = 105 \text{ min}$$